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Effects of different training modalities on lower-limb explosive power, acceleration, 20-m sprint performance, and change-of-direction ability in youth soccer players: a systematic review and network meta-analysis

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Abstract

Background Youth soccer players repeatedly perform explosive actions, short accelerations, linear sprints, decelerations, and multidirectional movements. However, the comparative effects of different structured physical-conditioning programmes remain uncertain.

Methods Seven databases were searched from inception to 3 July 2026 using a final expanded search strategy encompassing plyometric, strength or resistance, sprint, acceleration, speed, change-of-direction, neuromuscular, multicomponent, and combined training. Randomised controlled trials involving healthy youth soccer players were eligible. Intervention arms were classified using operational, content-based node definitions. Construct-restricted primary networks and expanded sensitivity networks were analysed using frequentist random-effects network meta-analysis. Hedges' adjusted g was preferentially calculated from post-intervention or final-follow-up means, standard deviations, and sample sizes. Estimates were presented so that positive values indicated better performance. P-scores were treated as descriptive ranking summaries. Risk of bias was assessed using an adapted study-level application of the five-domain RoB 2 framework, and confidence in the evidence was assessed using CINeMA. A post hoc strict-age sensitivity analysis excluded two age-boundary studies.

Results Eighty-nine studies were included in the expanded quantitative analysis, of which 74 contributed to at least one construct-restricted primary network. The primary lower-limb explosive-power, acceleration, 20-m sprint, and planned change-of-direction networks included 55, 20, 25, and 38 studies, respectively. Compared with usual soccer training, plyometric training combined with sprint and/or change-of-direction training showed favourable estimates for lower-limb explosive power (SMD 0.79, 95% CI 0.55 to 1.03), acceleration (1.19, 0.90 to 1.49), 20-m sprint

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performance (0.80, 0.33 to 1.28), and planned change-of-direction ability (1.46, 1.13 to 1.80). Corresponding I^2 values were 34.6%, 21.8%, 65.0%, and 41.0%. Between-design inconsistency was detected in the 20-m sprint ($P=0.0036$) and change-of-direction ($P=0.0007$) networks. CINeMA confidence for these four comparisons was low, low, very low, and low, respectively. Expanded sensitivity networks showed substantially greater heterogeneity. The highest-ranked intervention differed across outcome domains but remained consistent within each outcome across the three analysis sets. Excluding the two age-boundary studies did not materially alter the principal estimates.

Conclusions Plyometric training combined with sprint and/or planned change-of-direction training produced favourable comparative estimates across the four performance outcomes. However, evidence for several nodes and active-versus-active comparisons was sparse, heterogeneity in programmes and outcomes was present, inconsistency was detected in some networks, and confidence in the evidence was low or very low. These limitations do not support a conclusion that any training category is universally superior. The findings should be interpreted as provisional category-level signals rather than definitive training prescriptions.

Systematic review registration PROSPERO CRD420261347297, registered on 21 March 2026,https://www.crd.york.ac.uk/PROSPERO/view/CRD420261347297.

Keywords Soccer players, Network meta-analysis, Plyometric training, Sprint training, Change-of-direction ability

Introduction

Soccer is a high-intensity intermittent sport [1]. Youth players are required to transition rapidly between running, physical duels, deceleration, changes of direction, and re-acceleration. The ability to perform explosive and multidirectional actions contributes to performance in match situations such as evading defenders, contesting possession, initiating movement before shooting, and transitioning between attack and defence. In professional soccer, straight-line sprinting has been reported as particularly common in goal-related situations [2], whereas change-of-direction performance requires deceleration, redirection of movement, and subsequent re-acceleration [3, 4]. These qualities therefore represent important targets in the physical conditioning of youth soccer players.

In applied training settings, youth soccer programmes may not rely on a single isolated training method. Instead, they may combine several physical conditioning interventions, including plyometric training, strength training, sprint or change-of-direction training, and multicomponent neuromuscular training. These interventions may be implemented as adjuncts to regular soccer training and should be considered alongside technical–tactical practice, competition schedules, and the growth and maturity status of youth athletes [5, 6]. Depending on the modality, their primary training stimuli may target jumping ability, acceleration, deceleration and change of direction, force production, or broader neuromuscular adaptations.

Although previous systematic reviews and meta-analyses have reported improvements in selected performance outcomes among high-level male youth soccer players [6], recent network meta-analyses have also compared multiple training modalities in soccer populations [7–9]. However, differences in populations, intervention classifications, and outcome coverage across existing syntheses leave uncertainty regarding the relative effects of multiple structured training modalities across the four performance domains considered here. Network meta-analysis can integrate direct and indirect evidence and compare multiple intervention options simultaneously [10], making it suitable for addressing comparative questions involving several competing intervention strategies.

Therefore, this systematic review and network meta-analysis aimed to compare the relative effects of different structured training modalities on four domains of athletic performance in youth soccer players. In addition, the findings were interpreted in conjunction with risk-of-bias assessments and confidence-of-evidence evaluations to support cautious interpretation of the comparative evidence and identify priorities for future research.

Methods

Study design, registration, and revision-stage analytical decisions

This systematic review and network meta-analysis was registered in PROSPERO (CRD420261347297; registered on 21 March 2026). At the time of registration, the initial literature search had been completed, but title and abstract screening, full-text eligibility assessment, data extraction, and statistical analysis had not yet commenced. The search strategy was subsequently expanded during manuscript revision, after registration. The review was reported in accordance with PRISMA 2020 and the PRISMA extension for network meta-analysis [11, 12]. The registered review scope concerned the comparative effects of structured physical-training interventions on lower-limb explosive power, acceleration, sprint performance, and change-of-direction performance in youth soccer players.

This expansion was undertaken to improve coverage of all eligible intervention modalities. Before the revised

models were fitted, operational intervention-node definitions, an arm-level classification hierarchy, construct-restricted primary outcome networks, broader sensitivity networks, and comparison-level outcome-priority rules were formalised. A post hoc strict-age sensitivity analysis was also conducted by excluding two studies whose reported age ranges or distributions indicated that some participants were, or may have been, outside the prespecified 10–19-year range. These revision-stage decisions are explicitly distinguished from the original protocol specifications.

Literature search strategy

The expanded literature search was completed on 4 July 2026 and covered records from database inception to 3 July 2026. Seven electronic databases were searched: PubMed, Web of Science, Scopus, Cochrane Library, CINAHL, SPORTDiscus, and Embase. The final search strategy incorporated both the originally used plyometric/jump-related terms and the expanded terms required to capture all eligible intervention nodes in the network meta-analysis.

The search strategy was structured around four concept blocks: soccer players, youth populations, eligible training interventions, and randomised or controlled intervention study designs. Soccer-related terms included soccer, football, association football, soccer players, football players, youth soccer, academy soccer, and academy football. Youth-related terms included youth, adolescent, child, young, junior, teenage, academy, U13–U19, and under-13 to under-19. Intervention-related terms were expanded to cover all eligible training modalities, including plyometric or jump training, strength or resistance training, sprint, acceleration and speed training, change-of-direction or agility training, neuromuscular training, and multicomponent, combined, complex, or contrast training. Study-design terms included randomised, controlled trial, clinical trial, intervention, experimental, control group, comparison group, and intervention group.

The search strategy was adapted for each database according to its syntax and controlled vocabulary. The complete database-specific search strategies are provided in Supplementary File S1. The expanded search retrieved 3,935 records before duplicate removal. After duplicate removal, 1,929 records were retained for title and abstract screening.

Eligibility criteria

Eligible participants were healthy soccer players recruited from explicitly youth, academy, junior, prepubertal, or age-group teams, with a prespecified target age range of 10–19 years. Studies were eligible when the reported group mean or median age was within this range. Studies whose reported age ranges or dispersion measures indicated that some participants were, or may have been, outside this range were retained in the primary analysis only when the cohort was explicitly described as an organised youth soccer population and no separable adult subgroup was reported. These studies were classified as age-boundary cases and excluded in a post hoc strict-age sensitivity analysis.

Studies were excluded if they involved adult or professional adult soccer players, participants from sports other than soccer, purely technical or tactical training interventions, single-session acute interventions, interventions primarily designed for injury rehabilitation, or studies that did not provide sufficient data to calculate effect sizes. After eligibility assessment and outcome-network construction, 89 studies met the broad eligibility criteria for the expanded evidence set. Of these, 74 studies contributed to at least one construct-restricted primary network and were used for the main network meta-analysis. Studies that met the broader eligibility criteria but did not satisfy the construct-restricted primary network definitions were retained for expanded sensitivity analyses. Detailed PICOS information, participant characteristics, sample sizes, intervention duration, weekly training frequency, session duration, and outcome measures for the included studies are presented in Supplementary Tables S1A and S1B.

Intervention classification and outcome network specification

Interventions were classified at the intervention-arm level using operational rules based on the prescribed exercise content rather than solely on programme names used by the original investigators. CON comprised usual soccer training, routine team training, or no additional targeted physical-conditioning intervention. LST comprised programmes predominantly based on straight-line acceleration, maximal-speed sprinting, resisted linear sprinting, or repeated linear sprinting. CST comprised programmes predominantly based on sprinting along curved or arc-shaped trajectories. SCOD comprised stand-alone sprint and/or planned change-of-direction programmes, including repeated change-of-direction sprint training, that did not contain a substantive plyometric component and could not be adequately represented by LST or CST alone. PLY comprised programmes in which jumping, hopping, bounding, drop-jump, hurdle-jump, or other stretch–shortening-cycle exercises represented the predominant training stimulus. STR comprised programmes predominantly based on external-load, body-mass, maximal, submaximal, eccentric, flywheel, Nordic-hamstring, or other resistance-based strength exercises. STR + PLY comprised programmes containing separate and substantive strength and plyometric components. PLY+SCOD

comprised programmes combining substantive plyometric exercises with substantive sprint, acceleration, shuttle-sprint, agility, or planned change-of-direction exercises. MN comprised integrated programmes involving several co-dominant neuromuscular domains that could not be adequately represented by a narrower single- or two-component node, such as combinations of strength, core stability, balance, coordination, landing control, plyometrics, sprinting, and change-of-direction training.

A hierarchical classification rule was applied before the final revised network analyses. Programmes with one clearly predominant stimulus were assigned to the corresponding specific node. Programmes containing sprint or planned change-of-direction training without a substantive plyometric component were assigned to LST, CST, or SCOD according to the movement trajectory and prescribed training structure. Programmes with two co-dominant components were assigned to STR + PLY when the defining components were strength and plyometric training, or to PLY+SCOD when the defining components were plyometric and sprint/change-of-direction training. Programmes integrating three or more co-dominant neuromuscular domains, or programmes whose content could not be reduced to a narrower one- or two-component category, were assigned to MN. External resistance used only to increase the loading of a jump or sprint exercise did not by itself constitute a separate strength-training component. Regular technical–tactical soccer training performed by all study groups did not alter the intervention-node assignment.

Assignment to the same network node did not imply that training programmes were identical. It indicated that the programmes shared a predominant prescribed training stimulus and were sufficiently similar for the prespecified node classification. Differences in intervention duration, weekly frequency, session duration, jump type and volume, sprint distance, change-of-direction angle, external load, progression, and co-interventions were considered potential effect modifiers in the assessment of within-node heterogeneity and transitivity. Arm-level protocol content, dose, alternative plausible classifications, final node assignments, and classification rationales are reported in Supplementary Tables S1B and S2B, whereas operational node definitions, dose ranges, and principal within-node differences are summarised in Supplementary Table S2A.

Four outcome networks were constructed. N1 represented lower-limb explosive power. Direct vertical-jump height or horizontal-jump distance measures were prioritised. When several correlated explosive-performance tests were reported, one measure was selected according to the documented hierarchy. Squat jumps, drop jumps, derived power or velocity measures, and broader explosive-performance outcomes were restricted to the expanded sensitivity analysis or retained only as documented fallback measures when no higher-priority outcome was available.

N2 represented short-distance linear acceleration and was restricted in the construct-restricted primary analysis to 5-m, 10-m, or 0–10-m linear acceleration time. Physiological variables, agility or change-of-direction measures, COD deficits, reactive-agility outcomes, repeated-sprint indices, shuttle-COD outcomes, and other non-linear measures were not eligible for the primary N2 network. Broader short-distance linear speed measures were considered only under the expanded eligibility rules.

N3 represented complete 20-m linear sprint time. Split-only outcomes, non-standard sprint distances, curved sprint tests, repeated-sprint indices, and other broader speed-derived outcomes were excluded from the construct-restricted primary N3 network unless explicitly designated as sensitivity outcomes.

N4 represented planned change-of-direction ability. The original completion time for a single pre-planned course was prioritised over COD deficit, percentage loss, speed-derived outcomes, repeated-effort tests, and fatigue indices. A suffix such as “d” was not automatically interpreted as “deficit”. The measures 505-CODAd and COD10d were retained when full-text verification established that the suffix denoted the dominant side and that the measure represented an original route-completion time. Derived percentage-loss or deficit measures were otherwise restricted to expanded sensitivity analyses or the outcome-selection audit.

Within each study, outcome network, and assessment time point, only one outcome was selected for the construct-restricted primary analysis. Multiple highly correlated tests obtained from the same participants were not entered as independent comparisons. When the highest-priority measure was unavailable, a documented fallback measure could be retained; broader, derived, or lower-priority measures were otherwise restricted to the expanded sensitivity analysis.

Direction harmonisation ensured that effect estimates had a common statistical interpretation, with positive values representing better performance in the manuscript. However, direction harmonisation did not eliminate biomechanical differences between tests, including differences in jump technique, force orientation, sprint distance, change-of-direction angle, braking demand, movement route, and repeated-effort requirements.

Study selection and data extraction

Two reviewers (K.W. and J.H.) independently screened titles and abstracts and independently assessed potentially eligible full-text reports against the eligibility criteria. No formal prospective pilot calibration exercise was

conducted before screening. After both reviewers had completed their initial independent screening decisions, they undertook a structured post-screening calibration and reconciliation exercise to compare discrepant decisions, clarify the operational application of the eligibility criteria, and standardise the final eligibility decisions. Disagreements were resolved through discussion, and any unresolved eligibility decision was adjudicated by X.B. Reasons for full-text exclusion were recorded. Record-level selection counts were reconciled with the PRISMA flow diagram, whereas study-level membership of the primary and expanded sensitivity networks was reconciled separately using the canonical Study-ID mapping table.

Data were extracted and audited in a linked structure at five levels: study, intervention arm, outcome, assessment time point, and pairwise comparison. At the study level, the extracted variables included the canonical Study ID, full citation, publication year, country, study design, participant characteristics, sample size, age, sex, competitive level, and information relevant to multiple reports or possible sample overlap.

At the intervention-arm level, the original intervention label, comparator, intervention duration, weekly frequency, session duration, exercise components, training volume, sets, repetitions, contacts or running distance, loading strategy, progression method, co-interventions, adherence, adverse events or injuries where reported, and final network-node assignment were recorded. At the outcome and assessment-time-point levels, the original outcome label, measurement unit, direction of benefit, raw assessment time, standardised time-point code, assigned outcome network, and eligibility for the primary or expanded sensitivity analysis were recorded. At the comparison level, treatment nodes, sample sizes, means, standard deviations, effect-size basis, Hedges' adjusted g, standard error, direction-harmonisation status, and final model-input eligibility were documented.

K.W. completed the initial extraction, and J.H. independently checked the study characteristics and quantitative outcome data against the original reports. No formal prospective pilot calibration exercise was conducted before data extraction. After the initial extraction and independent checking had been completed, K.W. and J.H. conducted a structured calibration and reconciliation exercise to align the interpretation of the extraction fields and decision rules, recheck discrepant entries against the original reports, and correct the final dataset where necessary. Critical model-input fields, including group sample sizes, means, standard deviations, assessment times, outcome selection, treatment coding, and effect direction, were independently verified. A.Z. additionally verified intervention-node classification, training-dose information, assessment-time-point coding, canonical Study-ID mapping, and primary or expanded sensitivity network membership against the source reports and the explicit decision rules.

A single canonical Study ID was assigned to each study and was used consistently in the study-characteristics tables, arm-level datasets, RoB 2 files, model-input datasets, figures, and statistical outputs. The study_id and studlab fields were required to match the canonical Study ID. All comparisons from the same multi-arm trial retained one common study label and were not split into artificial study identifiers.

Potential duplicate publications, multiple reports, and possible participant overlap were evaluated using titles, DOIs, authorship, participant characteristics, sample sizes, intervention protocols, outcome batteries, recruitment settings, assessment schedules, and ethics information. No report was confirmed as a duplicate publication or secondary report of another included study; therefore, all 89 studies were retained as separate study units.

Assessment times were coded by combining the prefix "post_" with the numerical duration reported in the original study and the suffix "wk" or "mo" (e.g., post_8wk and post_3mo). Week- and month-based assessment times were retained in their originally reported units, and month-based times were not mechanically converted to weeks. The code post_end was reserved for assessments conducted at the end of the intervention when the report did not provide a sufficiently reliable numerical duration.

When discrepancies were identified between the initial extraction and the main text, tables, figure captions, appendices, or supplementary materials of an included report, the source report was rechecked. The final value and its page, table, figure, or appendix location were recorded. Information was coded as not reported only when it remained unavailable or could not be reliably recovered after full-text review; unreported intervention-dose values were not inferred from similar programmes reported in other studies.

Detailed study characteristics and intervention-dose data are presented in Supplementary Tables S1A and S1B. Original intervention-to-node mappings are presented in Supplementary Table S2B. Arm-level outcome data and final comparison-level model inputs are reported in Supplementary Tables S4A and S4B. Study-network membership, time-point standardisation, and special analytical decisions are documented in Supplementary Tables S6A–S6C. Reviewer responsibilities and verification procedures are reported in Supplementary Table S14.

Risk-of-bias assessment

Risk of bias was assessed at the study level through an adapted application of the five-domain Cochrane Risk of Bias 2 (RoB 2) framework [13]: bias arising from the

randomisation process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result.

K.W. and A.Z. independently evaluated each included study and recorded the supporting information for each domain-level and overall judgement. No formal prospective pilot calibration exercise was conducted before the assessments. After completing their initial independent evaluations, the two reviewers undertook a structured calibration and consensus process in which the signalling-question responses, supporting information, and domain-level and overall judgements were compared against the RoB 2 guidance. Disagreements were resolved through discussion and re-examination of the original report, protocol, registration record, tables, figures, and supplementary materials. Any unresolved judgement was adjudicated by X.B.

One consensus study-level judgement was retained for each included study. This adapted study-level application was used to provide one descriptive judgement for each trial and was not intended to constitute a formal outcome-result-level RoB 2 assessment. The study-level assessments were used for the descriptive risk-of-bias figures and narrative summary. Reviewer responsibilities and the independent assessment and consensus procedures are documented in Supplementary Table S14.

Statistical analysis and confidence-of-evidence assessment

Separate frequentist random-effects network meta-analyses were conducted for lower-limb explosive power, acceleration, 20-m sprint performance, and planned change-of-direction ability using the netmeta package in R [14]. Because eligible studies used different tests and measurement scales, the principal effect measure was the standardised mean difference expressed as Hedges' adjusted g with a 95% confidence interval. Means, standard deviations, and sample sizes at the prespecified post-intervention assessment time point were preferentially used. Post-intervention values were prioritised because baseline-adjusted estimates were inconsistently reported, change-score standard deviations were frequently unavailable, and deriving missing change-score standard deviations would have required assumptions regarding baseline–follow-up correlations that could not be applied consistently across studies. Although post-intervention and change-score SMDs can be combined under appropriate conditions, they were not combined within the same primary network to maintain a consistent effect-size basis [15]. The availability and usability of change-score and baseline-adjusted estimates were audited separately. Such estimates were eligible for a distinct sensitivity analysis only when they could be converted to an effect estimate and standard error and formed a sufficiently connected evidence network. Native model directions were retained for model fitting and treatment ranking. Larger model effects represented better performance for N1 and N2, whereas smaller native model effects represented better performance for N3 and N4. For publication tables and narrative reporting, the N3 and N4 effect estimates were multiplied by − 1, and their confidence limits were correspondingly reversed, so that positive SMDs consistently indicated better athletic performance. This presentation-only transformation did not alter standard errors, P values, Cochran's Q, I^2, τ^2, inconsistency statistics, or P-scores.

The construct-restricted primary networks were used for the principal conclusions. The expanded sensitivity networks were used to assess the robustness of the findings under broader study-, comparison-, and outcome-eligibility rules, whereas a separate post hoc strict-age sensitivity analysis was used to assess robustness to the inclusion of studies located at the prespecified participant-age boundaries. To limit non-independence arising from multiple correlated outcomes, the explicitly defined priority outcome was retained for each study, outcome network, and assessment time point in the primary analysis. Broader eligible outcomes were retained in the expanded sensitivity analyses.

The strict-age sensitivity analysis was derived directly from the final construct-restricted primary dataset. The two age-boundary studies identified in the canonical Study-ID mapping table as Abdelkader et al. (2022) and Chelly et al. (2010) were excluded as whole studies from all primary outcome networks to which they contributed. All other outcome-selection, assessment-time-point, intervention-node, effect-direction, network-connectivity checking, multi-arm handling, and random-effects modelling procedures were unchanged from the primary analysis. For each outcome, the numbers of studies, intervention nodes, and model-input contrasts, together with the SMDs and 95% CIs, Cochran's Q, I^2, τ^2, and random-effects between-design inconsistency statistics, were compared with the corresponding primary results.

Multi-arm studies retained a common canonical studlab, enabling netmeta to account for within-study correlation by reweighting the comparisons; tol.multiarm = 0.001 and tol.multiarm.se = 0.001 were used as numerical consistency tolerances for multi-arm treatment estimates and standard errors [14]. Multi-arm trials were not divided into artificial study labels. When two or more original intervention arms within the same study mapped to the same final network node, their arm-level sample sizes, means, and standard deviations were pooled before calculating a single node-level contrast. Such arms were not entered as separate independent comparisons. Original arms continued to be reported separately in the intervention-characteristics and arm-level data tables,

whereas the pooled node-level contrast was used only in the network model input.

Network geometry and connectivity were examined before model fitting. Statistical heterogeneity was evaluated using Cochran's Q, I^2, and τ^2. Random-effects relative-effect matrices, estimates versus CON, league tables, forest plots, and treatment rankings were generated. Random-effects P-scores were calculated using network-specific benefit directions [16]. Larger effects were specified as beneficial for N1 and N2, whereas smaller native model effects were specified as beneficial for N3 and N4. Rankings were treated as descriptive summaries rather than evidence of superiority and were interpreted only in conjunction with relative-effect estimates, confidence intervals, direct-evidence availability, network geometry, heterogeneity, inconsistency, risk of bias, and confidence in the evidence.

The transitivity assumption was evaluated by comparing potential effect modifiers across the available direct comparisons, including age, sex, competitive level, maturity status where reported, intervention duration, weekly frequency, session duration, training volume, loading strategy, progression, comparator content, outcome test, and assessment time point [17]. Global statistical inconsistency was assessed using random-effects design decomposition and the between-design Q statistic extracted from the corresponding final fitted model [18]. Local disagreement between direct and indirect evidence was examined using SIDE or node-splitting analyses where both evidence sources were available [19].

Potential small-study effects, including patterns compatible with reporting bias, were explored using comparison-adjusted funnel plots for the primary and expanded networks [20]. Funnel-plot patterns were interpreted cautiously because asymmetry could arise from heterogeneity, network structure, differences between comparisons, selective reporting, or chance. Egger regression was not performed, and the absence of publication bias was not inferred from visually symmetrical plots.

Confidence in each prespecified primary relative-effect estimate was assessed using CINeMA across six domains: within-study bias, reporting bias, indirectness, imprecision, heterogeneity, and incoherence [21]. An absolute SMD of 0.20 was prespecified as the threshold of practical importance for evaluating imprecision, heterogeneity, and incoherence [21]. Contribution matrices were used to determine how much information each included study contributed to each network estimate. RoB 2-linked study contributions, network estimates, direct estimates where available, confidence intervals, prediction intervals, heterogeneity statistics, random-effects design decomposition, and comparison-level SIDE results were used as supporting evidence. These statistical outputs supported but did not automatically determine the final domain judgements or overall confidence ratings.

K.W. and A.Z. independently assessed the six CINeMA domains for each prespecified primary comparison and independently assigned an initial overall confidence rating. No formal prospective pilot calibration exercise was conducted before the CINeMA assessments. After completing their initial independent assessments, the two reviewers undertook a structured post-assessment calibration and consensus exercise to harmonise the application of the six CINeMA domains, the prespecified absolute SMD threshold of 0.20, and the rules used to derive the overall confidence ratings. Differences in domain-level or overall judgements were resolved through discussion, with unresolved cases adjudicated by X.B. Final consensus domain judgements, reasons for downgrading, and confidence ratings are presented in Supplementary Table S13, whereas reviewer responsibilities and the independent assessment, calibration, consensus, and adjudication procedures are documented in Supplementary Table S14.

Results

Study selection and characteristics of the included studies

The expanded database search identified 3,935 records across the seven databases. After duplicate removal, 1,929 records were screened by title and abstract. Of these, 1,575 records were excluded, and 354 reports were sought for retrieval. Twenty-nine reports could not be retrieved, leaving 325 full-text reports for eligibility assessment. During full-text assessment, 236 reports were excluded for the reasons shown in Fig. 1. Eighty-nine studies were included in the systematic review and expanded quantitative analysis. Of these, 74 contributed to at least one construct-restricted primary network, whereas 15 contributed only to the expanded sensitivity networks.

The 74 studies contributing to at least one construct-restricted primary network were conducted across 17 countries: Tunisia (n = 25) [22–46], Spain (n = 15) [47–61], Türkiye (n = 7) [62–68], Greece (n = 5) [69–73], Chile (n = 4) [74–77], Iran (n = 3) [78–80], Norway (n = 3) [81–83], Denmark (n = 2) [84, 85], Portugal (n = 2) [86, 87], Canada (n = 1) [88], China (n = 1) [89], Colombia (n = 1) [90], India (n = 1) [91], Italy (n = 1) [92], Latvia (n = 1) [93], Sweden (n = 1) [94], and Uruguay (n = 1) [95]. The remaining 15 studies did not contribute to any construct-restricted primary network but entered at least one expanded sensitivity network. These expanded-network-only studies were conducted across nine countries: Türkiye (n = 4) [96–99], Italy (n = 3) [100–102], Tunisia (n = 2) [103, 104], China (n = 1) [105], the Czech Republic (n = 1) [106], Germany (n = 1) [107], Israel (n = 1) [108], Sweden (n = 1) [109], and the United States (n = 1)

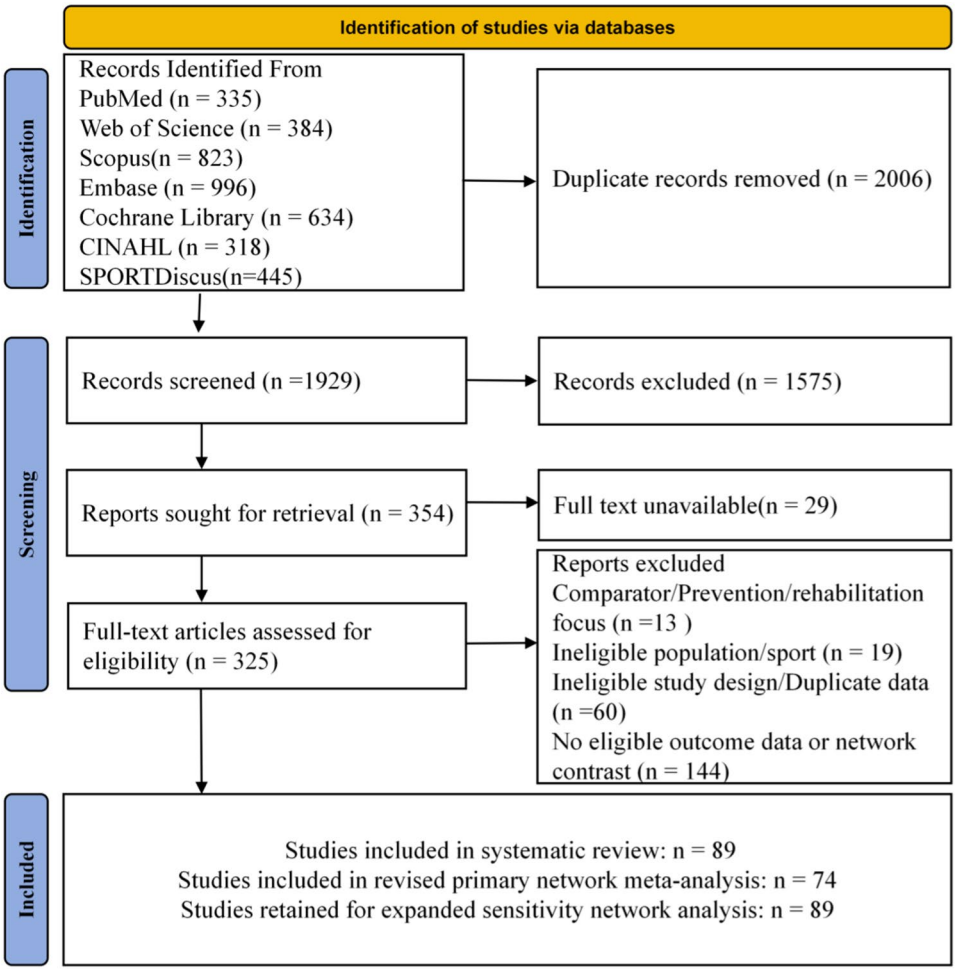


Fig. 1 PRISMA flow diagram for the final expanded search and study selection process

[110]. They met the broader review eligibility criteria but did not satisfy the construct-restricted primary-network definitions and therefore contributed exclusively to the expanded sensitivity analyses. Accordingly, no study conducted in the United States contributed to a construct-restricted primary network, although one United States study was retained in the expanded sensitivity evidence set. Country was defined according to the participant recruitment location or study setting rather than the institutional affiliation of the authors. Detailed study-level country and analysis-set information is provided in Supplementary Table S1A.

Potential duplicate publications or multiple reports were rechecked using titles, DOIs, sample sizes, participant characteristics, intervention protocols, outcome batteries, and canonical Study-ID mappings. No included study was identified as a confirmed duplicate or secondary report of another included study; therefore, all 89 were retained as separate study units.

Two studies, Abdelkader et al. (2022) and Chelly et al. (2010), were identified as age-boundary cases. Both were retained in the primary analysis under the organised-youth-cohort operational rule but were excluded as whole studies from the post hoc strict-age sensitivity analysis. The strict-age sensitivity analysis therefore included 72 studies.

Study-level characteristics, participant information, eligible outcomes, and network membership for all 89 studies are reported in Supplementary Table S1A. Intervention-arm training dose, implementation characteristics, source locations, and verification status are reported in Supplementary Table S1B. The complete primary, expanded sensitivity, and strict-age sensitivity network membership is reported in Supplementary Table S6A, and study-level age-boundary decisions and exclusions are reported in Supplementary Table S15B, whereas the corresponding network-level strict-age sensitivity results are reported in Supplementary Table S15A.

The primary analysis contained 74 unique studies and 156 final model-input contrast rows. The expanded sensitivity analysis contained 89 unique studies and 248 model-input contrast rows. The strict-age sensitivity

analysis contained 72 unique studies and 154 model-input contrast rows. The primary N1 lower-limb explosive-power network included 55 studies, eight nodes, and 63 model-input contrast rows. The N2 acceleration network included 20 studies, eight nodes, and 22 model-input contrast rows. The N3 20-m sprint network included 25 studies, six nodes, and 27 model-input contrast rows. The N4 planned change-of-direction network included 38 studies, eight nodes, and 44 model-input contrast rows. The expanded N1, N2, N3, and N4 networks included 65, 39, 37, and 65 studies; eight, nine, eight, and nine nodes; and 75, 47, 47, and 79 model-input contrast rows, respectively. Study counts are network specific and should not be summed across outcomes because one study could contribute to more than one network, as summarised in Table 1.

Risk of bias

Risk of bias was assessed at the study level through an adapted application of the five-domain RoB 2 framework [13]. Among the 74 studies contributing to at least one construct-restricted primary network, 72 were judged to have some concerns and two were judged to be at high risk of bias. Across all 89 studies included in the expanded evidence set, 87 were judged to have some concerns and two were judged to be at high risk of bias. No study was judged to be at low risk of bias.

Figures 2 and 3 present the domain-level and overall study-level risk-of-bias judgements for the 74 studies contributing to at least one construct-restricted primary network. The most frequent concerns related to the randomisation process and selection of the reported result, particularly where allocation procedures, prospective trial registration, published or accessible protocols, or prespecified analysis plans were insufficiently reported. The consensus study-level risk-of-bias judgement for each study was considered together with the study's proportional contribution to each network estimate when informing the within-study-bias domain of the CINeMA evaluations [21].

The structures of the four construct-restricted primary networks are shown in Fig. 4. The N1 lower-limb explosive-power network included 55 studies, 63 model-input contrast rows, and eight intervention nodes. The N2 acceleration network included 20 studies, 22 model-input contrast rows, and eight nodes. The N3 20-m sprint network included 25 studies, 27 model-input contrast rows, and six nodes. The N4 planned change-of-direction network included 38 studies, 44 model-input contrast rows, and eight nodes. Individual studies could contribute to more than one outcome network; therefore, study counts should not be summed across networks.

Network meta-analysis results

The N1 lower-limb explosive-power network included 55 studies, eight nodes, and 63 model-input contrast rows. PLY+SCOD versus CON yielded an SMD of 0.788 (95% CI 0.545 to 1.030; $P < 0.0001$). Between-study heterogeneity was statistically detectable (Q = 79.50, df = 52, $P = 0.0084$; $I^2 = 34.6\%$; $\tau^2 = 0.0716$), and the random-effects between-design test did not identify a statistically detectable global inconsistency signal ($P = 0.7057$). MN had the highest descriptive P-score (0.855), followed by STR + PLY (0.811), whereas PLY+SCOD ranked third (0.784) [16]. The MN-versus-CON estimate was 0.863 (95% CI 0.574 to 1.152); therefore, PLY+SCOD was not the highest-ranked intervention in N1.

The N2 acceleration network included 20 studies, eight nodes, and 22 model-input contrast rows. PLY+SCOD versus CON yielded an SMD of 1.195 (95% CI 0.904 to 1.486; $P < 0.0001$). Between-study heterogeneity was not statistically detectable (Q = 17.90, df = 14, $P = 0.2114$; $I^2 = 21.8\%$; $\tau^2 = 0.0371$), and the random-effects between-design P value was 0.3096. PLY+SCOD had the highest descriptive P-score (0.962).

The N3 20-m sprint network included 25 studies, six nodes, and 27 model-input contrast rows. After

Table 1 Composition of the construct-restricted primary and expanded sensitivity networks

Outcome	Analysis set	Studies, *n*	Nodes, *n*	Model-input contrast rows, *n*
Lower-limb explosive power	Primary	55	8	63
Acceleration	Primary	20	8	22
20-m sprint performance	Primary	25	6	27
Planned change-of-direction ability	Primary	38	8	44
Lower-limb explosive power	Expanded sensitivity	65	8	75
Acceleration	Expanded sensitivity	39	9	47
20-m sprint performance	Expanded sensitivity	37	8	47
Planned change-of-direction ability	Expanded sensitivity	65	9	79

The primary analysis included 74 unique studies and 156 model-input contrast rows, whereas the expanded sensitivity analysis included 89 unique studies and 248 contrast rows. Study counts are unique within each outcome network and are not additive because individual studies could contribute to more than one outcome network. Counts were generated after canonical Study-ID mapping, outcome-priority selection, time-point auditing, same-node arm pooling, construct validation, and comparison-level exclusions. The post hoc strict-age sensitivity analysis included 72 studies and 154 model-input contrast rows and is reported in Supplementary Table S15A

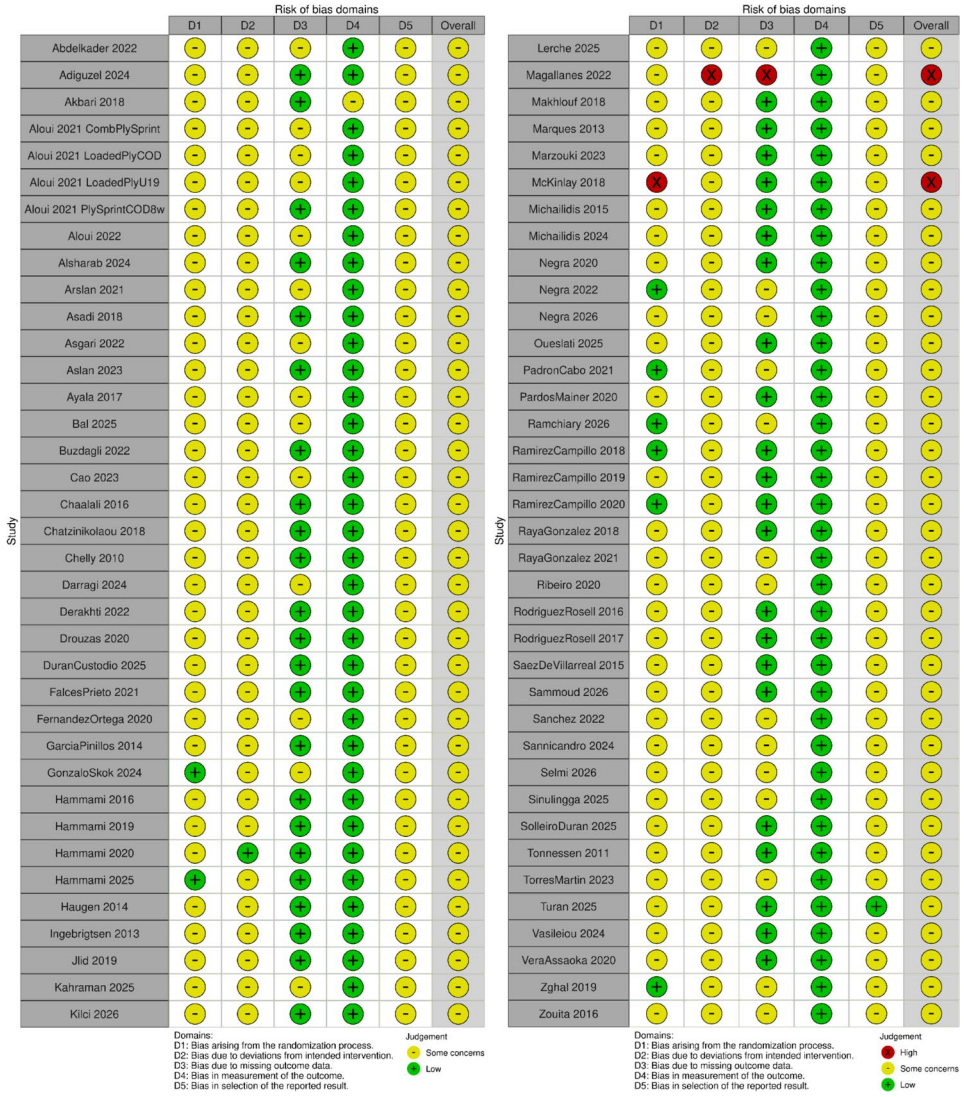


Fig. 2 Study-level risk-of-bias judgements for the 74 studies contributing to at least one construct-restricted primary network. Note: The traffic-light plot presents consensus study-level risk-of-bias judgements derived from an adapted study-level application of the five-domain RoB 2 framework for the 74 studies contributing to at least one construct-restricted primary network. These judgements represent descriptive study-level assessments and should not be interpreted as formal outcome-result-level RoB 2 assessments. One judgement was recorded for each study in each of the five domains and for the overall risk of bias. Green indicates low risk of bias, yellow indicates some concerns, and red indicates high risk of bias. D1, bias arising from the randomisation process; D2, bias due to deviations from intended interventions; D3, bias due to missing outcome data; D4, bias in measurement of the outcome; D5, bias in selection of the reported result

presentation-direction harmonisation, PLY+SCOD versus CON yielded an SMD of 0.802 (95% CI 0.326 to 1.278; $P = 0.001$). Between-study heterogeneity was statistically detectable ($Q = 59.97$, $df = 21$, $P < 0.0001$; $I^2 = 65.0\%$; $\tau^2 = 0.2241$), and statistically detectable between-design inconsistency was present ($P = 0.0036$). PLY+SCOD had the highest descriptive P-score (0.893).

The N4 planned change-of-direction network included 38 studies, eight nodes, and 44 model-input contrast rows. After presentation-direction harmonisation, PLY+SCOD versus CON yielded an SMD of 1.464 (95% CI 1.128 to 1.800; $P < 0.0001$). Between-study heterogeneity was statistically detectable ($Q = 57.62$, $df = 34$, $P = 0.0069$; $I^2 = 41.0\%$; $\tau^2 = 0.0928$), and statistically detectable between-design inconsistency was present ($P = 0.0007$). PLY+SCOD had the highest descriptive P-score (0.932).

Across the four primary networks, PLY+SCOD showed favourable estimates relative to CON. However, this finding should not be interpreted as evidence that PLY+SCOD was uniformly superior to all active interventions. MN had the highest descriptive ranking and a numerically larger estimate than PLY+SCOD in N1, while several active-versus-active comparisons were imprecise.

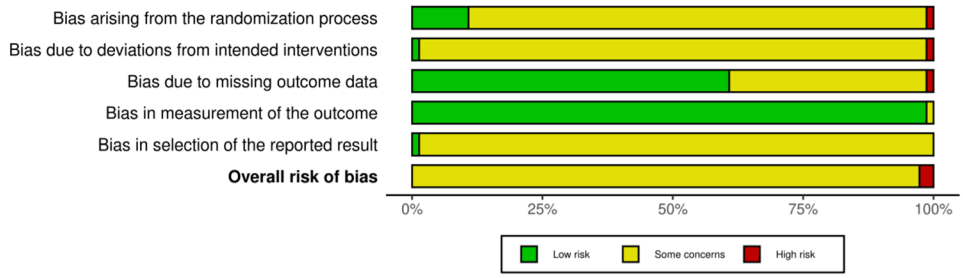


Fig. 3 Distribution of study-level risk-of-bias judgements across the 74 studies contributing to at least one construct-restricted primary network. Note: The bar plot presents the percentage distribution of consensus study-level risk-of-bias judgements derived from an adapted study-level application of the five-domain RoB 2 framework across the 74 studies contributing to at least one construct-restricted primary network. These distributions represent descriptive study-level summaries and should not be interpreted as formal outcome-result-level RoB 2 summaries. Each study was counted once and was not weighted by sample size or by the number of outcome networks to which it contributed. Green indicates low risk of bias, yellow indicates some concerns, and red indicates high risk of bias. D1, bias arising from the randomisation process; D2, bias due to deviations from intended interventions; D3, bias due to missing outcome data; D4, bias in measurement of the outcome; D5, bias in selection of the reported result

CINeMA confidence for the PLY+SCOD-versus-CON comparisons was low for N1, low for N2, very low for N3, and low for N4 Table 2.

In addition to the comparison between PLY+SCOD and regular soccer training, the direction of the relative effect estimates for strength training (STR) and strength combined with plyometric training (STR + PLY) was not fully consistent across most outcome networks, and the 95% confidence intervals of several comparisons crossed 0. Given the low or very low CINeMA confidence for these comparisons, STR and STR + PLY were not interpreted as training modes with definitive effects. Instead, they were treated as exploratory signals, with favourable trends in some comparisons. Detailed comparison effect sizes, contributions of direct and indirect evidence, inconsistency assessments, and confidence-of-evidence ratings are provided in the Supplementary Material.

Sensitivity analyses, heterogeneity, inconsistency, reporting bias, and confidence in the evidence

The expanded N1, N2, N3, and N4 networks included 65, 39, 37, and 65 studies and 75, 47, 47, and 79 model-input contrast rows, respectively. Corresponding I^2 values were 68.9%, 79.8%, 73.7%, and 80.0%, indicating substantially greater observed heterogeneity under the broader eligibility rules. Random-effects between-design P values were 0.8730, 0.5970, 0.5514, and 0.0012, respectively. A non-significant inconsistency test was not interpreted as evidence that the network was fully coherent [17, 114, 117].

After presentation-direction harmonisation, the expanded PLY+SCOD-versus-CON estimates were 0.873 (95% CI 0.526 to 1.220) for N1, 0.939 (0.481 to 1.398) for N2, 0.809 (0.306 to 1.311) for N3, and 1.049 (0.570 to 1.527) for N4. Although all four point estimates remained favourable, their magnitudes, uncertainty, and heterogeneity differed from the construct-restricted primary results.

The highest-ranked intervention differed across outcome domains but remained consistent within each outcome across the three analysis sets. MN had the highest P-score in N1, whereas PLY+SCOD had the highest P-score in N2, N3, and N4 in the primary, expanded, and strict-age analyses. In the expanded analyses, the corresponding P-scores were 0.803, 0.909, 0.904, and 0.788, respectively. These rankings were treated as descriptive summaries rather than evidence of intervention superiority.

No statistically detectable local SIDE signal was identified in the primary networks. Borderline local signals were observed for MN versus PLY+SCOD in primary N1 ($P = 0.0654$) and PLY versus CON in primary N3 ($P = 0.0914$). In expanded N4, statistically detectable local disagreement was observed for MN versus CON ($P = 0.0001$), STR + PLY versus CON ($P = 0.0247$), MN versus STR + PLY ($P < 0.0001$), and STR versus STR + PLY ($P = 0.0049$). These findings were interpreted as comparison-specific incoherence signals and supported cautious interpretation of the affected network. Complete SIDE results are reported in Supplementary Table S11.

The post hoc strict-age sensitivity analysis included 72 studies and 154 model-input contrast rows. N1 included 54 studies, eight nodes, and 62 contrasts; N2 included 20 studies, eight nodes, and 22 contrasts; N3 included 25 studies, six nodes, and 27 contrasts; and N4 included 37 studies, eight nodes, and 43 contrasts. The PLY+SCOD-versus-CON estimate changed from 0.788 (95% CI 0.545 to 1.030) to 0.789 (0.545 to 1.033) in N1 and from 1.464 (1.128 to 1.800) to 1.466 (1.125 to 1.808) in N4. N2 and N3 were unchanged because neither age-boundary study contributed to those primary networks. The highest-ranked categories were unchanged in the strict-age analysis: MN ranked highest in N1, whereas PLY+SCOD ranked highest in N2, N3, and N4.

CINeMA assessment covered 26 designated primary comparisons. N1 included six low- and one

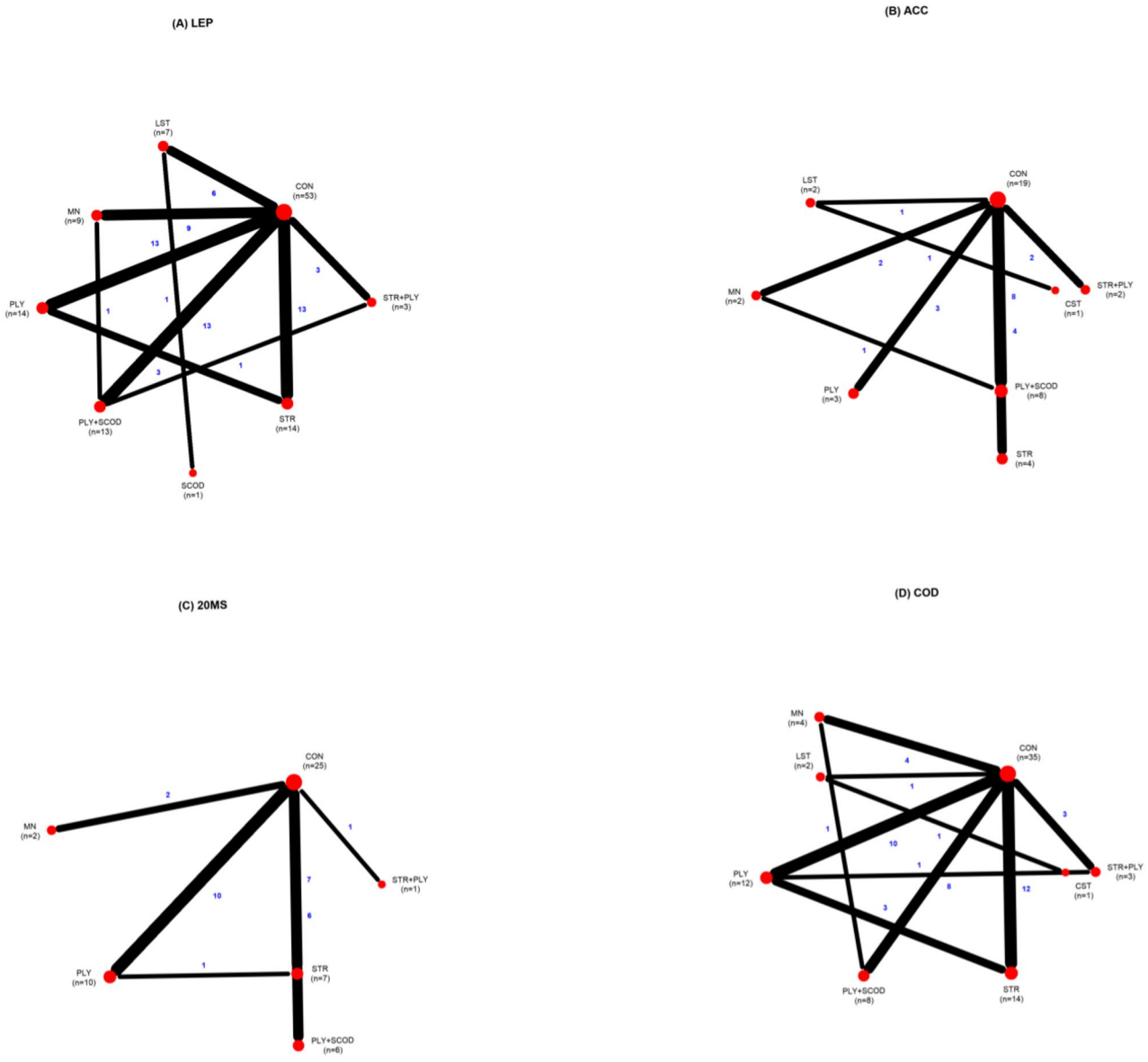


Fig. 4 Network evidence structures for the four construct-restricted primary outcome networks. Note: Nodes represent intervention categories, and lines represent direct comparisons. Node size and line thickness indicate the amount of available evidence. The N1, N2, N3, and N4 networks included 55, 20, 25, and 38 studies and 63, 22, 27, and 44 model-input contrast rows, respectively. LEP, lower-limb explosive power; ACC, acceleration; 20MS, 20-m sprint performance; COD, planned change-of-direction ability; CON, control or usual soccer training; LST, linear sprint training; CST, curved sprint training; SCOD, stand-alone sprint and/or planned change-of-direction training; MN, multicomponent neuromuscular training; PLY, plyometric training; PLY+SCOD, plyometric training combined with sprint and/or planned change-of-direction training; STR, strength or resistance training; STR+PLY, strength combined with plyometric training

very-low-confidence comparison; N2 included five low- and two very-low-confidence comparisons; all five N3 comparisons were rated very low; and N4 included four low- and three very-low-confidence comparisons. For PLY+SCOD versus CON, confidence was low for N1, low for N2, very low for N3, and low for N4.

Comparison-adjusted funnel plots were interpreted as exploratory graphical assessments [20, 111, 118]. Egger regression was not performed, and no inference that publication or selective-reporting bias was absent was based on apparent funnel-plot symmetry.

Discussion

Principal findings and interpretation

The construct-restricted primary analyses identified favourable PLY+SCOD-versus-CON estimates for all four performance outcomes. The highest-ranked intervention differed across outcome domains but was consistent within each domain across the primary, expanded,

Table 2 Summary of the primary network meta-analysis findings

Outcome	Studies, *n*	Nodes, *n*	Model-input contrasts, *n*	Key comparison	SMD (95% CI)	*P* for effect	Q (df)	I^2, %	τ^2	Design *P*	CINeMA confidence
LEP	55	8	63	PLY+SCOD vs. CON	0.788 (0.545 to 1.030)	< 0.0001	79.50 (52)	34.6%	0.0716	0.7057	Low
ACC	20	8	22	PLY+SCOD vs. CON	1.195 (0.904 to 1.486)	< 0.0001	17.90 (14)	21.8%	0.0371	0.3096	Low
20MS	25	6	27	PLY+SCOD vs. CON	0.802 (0.326 to 1.278)	0.001	59.97 (21)	65%	0.2241	0.0036	Very low
COD	38	8	44	PLY+SCOD vs. CON	1.464 (1.128 to 1.800)	< 0.0001	57.62 (34)	41%	0.0928	0.0007	Low

Positive SMDs favour the intervention listed first and indicate better athletic performance. Native N3 and N4 model estimates were oriented so that smaller values represented better performance; their SMDs and confidence-interval limits were multiplied by −1 for manuscript presentation only

CINeMA Confidence refers to the comparison-specific assessment for PLY+SCOD versus CON, *LEP* Lower-limb explosive power, *ACC* Acceleration, *20MS* 20-m sprint performance, *COD* Planned change-of-direction ability, *CON* Control or usual soccer training, *PLY+SCOD* Plyometric training combined with sprint and/or planned change-of-direction training, *SMD* Standardised mean difference, *CI* Confidence interval

This sign transformation did not alter standard errors, *P* values, Q, I^2, τ^2, between-design inconsistency statistics, or P-scores. Design P is the random-effects between-design inconsistency *P* value extracted from the corresponding design-decomposition model

and strict-age analyses. MN ranked highest for lower-limb explosive power, whereas PLY+SCOD ranked highest for acceleration, 20-m sprint performance, and planned change-of-direction ability. In primary N1, MN also had a numerically larger estimate versus CON than PLY+SCOD. These rankings remained descriptive and should not be interpreted as evidence of intervention superiority [16]. Moreover, CINeMA confidence in the principal PLY+SCOD-versus-CON comparisons was low or very low, which further limits the strength of any comparative conclusion [21].

From a training perspective, the favourable PLY+SCOD-versus-CON estimates may be consistent with combined exposure to rapid stretch–shortening-cycle actions, short-distance acceleration, braking, re-acceleration, and planned directional changes [112, 113]. Such programmes include several explosive movement tasks that are relevant to soccer, including jumping, acceleration, braking, re-acceleration, and planned directional changes [100]. However, the included studies did not directly measure neuromuscular recruitment, force–time characteristics, mechanical output, or the mechanisms underlying transfer to sport-specific performance. This explanation should therefore be regarded as a plausible training rationale rather than a causal mechanism demonstrated by the present network meta-analysis.

The distinction between PLY+SCOD and MN was based on the structure and relative contribution of the prescribed training components rather than on programme labels or the presence of an isolated exercise. PLY+SCOD required both a substantive plyometric component and a substantive sprint, acceleration, or planned change-of-direction component, without a separate co-dominant strength block or a broader multicomponent neuromuscular structure. MN was reserved for integrated programmes containing several co-dominant neuromuscular domains that could not be adequately represented by a narrower single- or two-component category. Accordingly, MN should not be interpreted as one standardised intervention, but as a category of integrated programmes sharing a broadly comparable multicomponent training rationale.

SCOD represented stand-alone sprint and/or planned change-of-direction training without a substantive plyometric component. Accordingly, repeated change-of-direction sprint training was not classified as PLY+SCOD merely because sprinting, braking, or directional changes involved stretch–shortening-cycle activity. Classification depended on the prescribed structure of the training programme rather than on biomechanical features that occur incidentally during the exercise.

Interventions assigned to the same node were not identical in training dose or exercise prescription. Programmes differed in intervention duration, weekly frequency, session duration, jump contacts, sprint distance, change-of-direction angle, external loading, progression, and exposure to regular soccer training. These characteristics were considered potential effect modifiers when assessing transitivity and within-node heterogeneity [114]. Nevertheless, incomplete reporting and residual within-node variation mean that each network estimate represents an average effect for a training category rather than the effect of one standardised protocol. The substantially greater heterogeneity observed in the expanded networks further suggests that the findings were sensitive to how intervention, comparison, and outcome boundaries were defined.

The findings for STR and STR + PLY require similarly cautious interpretation. Although these interventions may theoretically support force production, braking control, and re-acceleration [4], most relevant comparisons were imprecise, and several confidence intervals crossed

the null value. The confidence of evidence was also predominantly low or very low. Consequently, the present findings indicate possible favourable signals for selected outcomes but do not establish clear comparative advantages of STR or STR + PLY.

Although jumping, acceleration, 20-m linear sprinting, and planned change-of-direction performance are all explosive performance outcomes, the testing tasks and training adaptations underlying these outcomes are not identical [4, 112, 115]. Accordingly, the favourable PLY+SCOD-versus-CON estimates across the four primary networks should be interpreted as outcome-specific comparative signals rather than as evidence of an absolute advantage across all performance dimensions, all youth soccer players, or all training contexts.

With respect to specific outcomes, jumping performance is influenced by stretch–shortening-cycle function [112]. Acceleration and linear sprint performance are influenced by horizontal force application and sprint mechanical characteristics [115]. Planned COD performance additionally involves braking, control of the body's centre of mass, transition of support, and re-acceleration ability [4]. At the same time, change-of-direction ability in the present study refers only to COD tests performed under pre-planned route conditions. It does not include perception, decision-making, or responses to external stimuli in open match-play contexts [3]. Therefore, the findings should not be directly extrapolated to improvements in reactive agility [3].

Effect-direction harmonisation resolved whether higher or lower numerical values represented improvement, but it did not make different performance tests biomechanically interchangeable. Vertical jumps, horizontal jumps, drop jumps, straight-line acceleration, complete 20-m sprinting, 505 tests, shuttle routes, and other planned COD tests differ in force orientation, braking requirements, re-acceleration demands, movement routes, and measurement properties. Moreover, some retained fallback measures represented broader constructs than the highest-priority tests. The construct-restricted networks were therefore given greater interpretive weight than the expanded sensitivity networks.

Global between-design inconsistency was statistically detectable in the primary N3 and N4 networks. However, no statistically detectable local SIDE signal was identified in the primary networks. Borderline local signals were observed for MN versus PLY+SCOD in primary N1 and PLY versus CON in primary N3. Statistically detectable local direct–indirect disagreement was observed for several comparisons in expanded N4. SIDE estimates were unavailable for some comparisons when direct and indirect evidence were not both present, and other estimates were imprecise when evidence was sparse. This reduced the ability of global and local inconsistency procedures to identify disagreement reliably [114, 117].

Practical implications

In applied settings, programmes combining substantive plyometric and short-distance sprint or planned change-of-direction components may be considered when the objective is to address several explosive and speed-related qualities within the same conditioning programme. However, the present estimates represent average effects for broad training categories rather than validated prescriptions for one standardised programme. The findings do not establish that PLY+SCOD should replace strength, technical–tactical, or other conditioning components, nor do they establish one universally optimal training modality.

Training selection should account for player maturity, training history, technical competence, competition schedule, regular soccer workload, intervention dose, available supervision, and the specific physical quality being targeted [5, 116]. STR, PLY, STR + PLY, SCOD, and MN may remain relevant under different training objectives and periodisation contexts, but the available evidence does not support a precise hierarchy for programme prescription. Coaches should therefore use the present findings as one component of decision-making rather than as stand-alone instructions for training design.

Study limitations

Limitations should be considered. First, the evidence networks were uneven, and some nodes or active-versus-active comparisons were informed by few studies, limited direct evidence, or imprecise estimates. This reduced the ability of global and local inconsistency procedures to identify disagreement reliably [114, 117]. Statistically detectable between-design inconsistency was present in the primary N3 and N4 networks and in expanded N4. No statistically detectable local SIDE signal was identified in the primary networks, although borderline local signals were observed in primary N1 and N3; several expanded N4 comparisons showed statistically detectable local direct–indirect disagreement. Conversely, the absence of a statistically significant inconsistency test in other networks should not be interpreted as proof of coherence [114, 117].

Broader study, comparison, and outcome eligibility substantially increased heterogeneity. The expanded networks provide useful sensitivity analyses of broader evidence boundaries, but they should not be interpreted as equally construct-specific or more reliable alternatives to the primary networks. Although the complete ordering and P-score magnitudes of lower-ranked interventions varied across the primary and expanded analyses, the

highest-ranked category within each outcome remained unchanged. Rankings nevertheless remained descriptive because the expanded networks showed substantially greater heterogeneity and several relative-effect estimates were imprecise.

Interventions grouped within the same node differed in duration, weekly frequency, session duration, exercise selection, training volume, jump contacts, sprint distance, change-of-direction angle, external loading, progression, adherence, and co-interventions. Participant maturity, training age, baseline fitness, and exposure to regular soccer training were also incompletely reported. Although these variables were re-audited and considered as potential effect modifiers, incomplete reporting limited the assessment of transitivity and residual within-node heterogeneity. The operational hierarchy improved the reproducibility of node assignment but did not make all programmes within a node clinically homogeneous.

Effect-direction harmonisation did not remove biomechanical differences between outcome tests. The one-outcome-per-study hierarchy reduced repeated use of highly correlated measures obtained from the same participants, but the selected tests still differed in movement pattern, force orientation, distance, braking demand, direction-change angle, repeated-effort requirements, and measurement properties. Some lower-priority or broader outcomes were retained as documented fallback measures when a higher-priority outcome was unavailable. Consequently, each network estimate should be interpreted as an average across related but not identical tests.

The primary analyses preferentially used post-intervention or final-follow-up values. Post-intervention and change-score SMDs were not combined in the same primary network because change-score standard deviations and baseline-adjusted estimates were inconsistently available. A separate connected evidence network based exclusively on change scores or baseline-adjusted effects was not established in the present analysis. Sensitivity to the effect-data type therefore remains uncertain, and robustness was not inferred from unavailable alternative-effect estimates.

Participant-age eligibility was determined from aggregate study reports rather than individual participant data. Two studies were retained under the organised-youth-cohort operational rule because the cohorts were explicitly described as youth soccer populations and no separable adult subgroup was reported, although their aggregate age data indicated that some participants were, or may have been, outside the prespecified 10–19-year range. Their exclusion produced only minimal changes in the principal estimates, but the strict-age analysis was post hoc and cannot fully resolve uncertainty regarding individual-level age eligibility.

The evidence base was dominated by male participants, and many trials included relatively small samples. The primary-network evidence was also geographically concentrated, with 25 of the 74 studies conducted in Tunisia and 15 in Spain. This concentration in a limited number of settings may reduce the geographical and contextual generalisability of the findings and, where multiple trials originated from the same research groups, may introduce correlated design, implementation, or reporting practices across nominally separate studies. Reporting of training adherence, intervention-related adverse events, and injuries was limited and inconsistent, precluding firm conclusions regarding programme feasibility and safety. Non-reporting was not interpreted as evidence that no adverse events or injuries occurred.

Reporting-bias assessment was limited by sparse networks, differences in study size, heterogeneous outcome reporting, and limited protocol or registration availability. Comparison-adjusted funnel plots were interpreted only as exploratory graphical assessments [20, 111, 118]. Egger regression was not performed, and visually symmetrical funnel plots were not interpreted as evidence that publication or selective-reporting bias was absent [118].

No formal prospective pilot calibration was conducted before screening, data-extraction verification, RoB 2 assessment, or CINeMA assessment. However, after the relevant reviewers had completed their initial independent work, structured calibration and reconciliation were undertaken before the final eligibility decisions, datasets, and methodological judgements were locked, with any unresolved disagreements adjudicated by X.B.

Because a single study-level judgement was retained for each trial, this adapted assessment may not fully capture variation in risk of bias across different outcome results contributed by the same study. The study-level summaries should therefore be interpreted as descriptive review-level assessments rather than formal outcome-result-level RoB 2 judgements.

Most RoB 2 assessments indicated some concerns, and CINeMA confidence in the principal comparisons was low or very low. CINeMA judgements incorporated within-study bias, reporting bias, indirectness, imprecision, heterogeneity, and incoherence; therefore, favourable point estimates or high P-scores should not be interpreted independently of these concerns [13, 16, 21]. Treatment rankings and favourable estimates should be regarded as uncertain category-level signals rather than definitive evidence that one intervention was superior.

Conclusions

Plyometric training combined with sprint and/or planned change-of-direction training showed favourable comparative estimates for lower-limb explosive power,

acceleration, 20-m sprint performance, and planned change-of-direction ability in youth soccer players. However, PLY+SCOD did not occupy the highest descriptive rank for lower-limb explosive power, for which MN ranked highest across the primary, expanded, and strict-age analyses; PLY+SCOD ranked highest for acceleration, 20-m sprint performance, and planned change-of-direction ability. Confidence in the principal estimates remained low or very low. The findings should therefore be interpreted as provisional category-level signals rather than evidence that one training modality is universally superior or as prescriptive training recommendations. Future trials should preregister their protocols, report intervention content and training dose transparently, use clearly defined and construct-specific outcomes, and provide sufficient post-intervention, change-score, or baseline-adjusted data for independent synthesis.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13102-026-02041-9.

Additional file 1: Supplementary checklists, methods, tables, and figures. This file contains detailed study characteristics, intervention-node definitions, outcome-network mapping, effect-size calculations, primary and expanded network-construction rules, strict-age sensitivity analyses, reviewer-process documentation, CINeMA consensus assessments, relative-effect matrices, ranking results, inconsistency assessments, and supplementary figures.

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Authors' contributions
K.W. and X.B. conceived and designed the study. K.W. developed the search strategy, performed the literature search, screened studies, extracted data, conducted the statistical analyses, and drafted the original manuscript. J.H. assisted with study screening, data extraction, intervention classification, and preparation of tables and supplementary materials. A.Z. contributed to risk-of-bias assessment, certainty-of-evidence assessment, data verification, and interpretation of the results. X.B. supervised the study, provided methodological guidance, contributed to interpretation of the findings, and critically revised the manuscript for important intellectual content. All authors reviewed and approved the final manuscript.

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Data availability
The data that support the findings of this study are available within the article and its supplementary materials.

Declarations

Ethics approval and consent to participate
Not applicable. This study is a systematic review and network meta-analysis of previously published studies and did not involve the recruitment of new human participants, the collection of new human data, or any intervention involving human participants.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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